WORKED SOLUTION

STEP III 2010, Question 12

Probability and Statistics • Geometric Distribution Expectation

STEP III 2010, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q12

The key idea

The series S is an arithmetico-geometric series, and the trick of subtracting rS turns it into an ordinary geometric series. Once we have it, every expectation in the question is a sum of exactly this shape: the shot number grows linearly while the probability of getting that far shrinks geometrically. So the whole question comes down to writing each expectation as a sum and reading off the right values of r and d.

Solution

The series

Since $|r| < 1$, the series converges absolutely (its terms are bounded by a constant times $n|r|^n$), so we may subtract term by term. Lining up equal powers of r,

$$\begin{aligned} S &= 1 + (1+d)r + (1+2d)r^2 + (1+3d)r^3 + \cdots, \\ rS &= \qquad r + (1+d)r^2 + (1+2d)r^3 + \cdots, \end{aligned}$$

and each coefficient of r^n with $n \geq 1$ differs by exactly d. Hence

$$(1-r)S = 1 + d\left(r + r^2 + r^3 + \cdots\right) = 1 + \frac{rd}{1-r},$$

summing the geometric series, which is valid as $|r| < 1$. Dividing by $1 - r \neq 0$,

$$S = \frac{1}{1-r} + \frac{rd}{(1-r)^2},$$

as required.

Arthur alone

We assume $0 < a \leq 1$ (if $a = 0$ Arthur never hits). Let N be the number of shots Arthur needs. Then $N = n$ exactly when his first $n-1$ shots miss and the nth hits, so by

independence

$$\mathrm{P}(N = n) = a'^{n-1}a, \qquad n = 1, 2, 3, \ldots$$

Therefore

$$\mathrm{E}(N) = \sum_{n=1}^{\infty} n\, a\, a'^{n-1} = a(1 + 2a' + 3a'^2 + \cdots).$$

The bracket is S with $r = a'$ and $d = 1$ (the coefficient of a'^n is $1 + n$), and $0 \leq a' < 1$, so

$$\mathrm{E}(N) = a\left(\frac{1}{1-a'} + \frac{a'}{(1-a')^2}\right) = a\left(\frac{1}{a} + \frac{a'}{a^2}\right) = \frac{a + a'}{a} = \frac{1}{a},$$

using $1 - a' = a$ and then $a + a' = 1$. (This formula also gives 1 when $a = 1$.)

The contest: who wins

From now on take $0 < a < 1$ and $0 < b < 1$, so that $0 < a'b' < 1$. Shots $1, 3, 5, \ldots$ are Arthur's and shots $2, 4, 6, \ldots$ are Boadicea's. Call one Arthur shot followed by one Boadicea shot a *round*; a round passes with nobody hitting with probability $a'b'$.

Arthur wins on shot $2k + 1$ ($k = 0, 1, 2, \ldots$) exactly when the first k rounds pass with no hit and then Arthur hits. By independence this has probability $(a'b')^k a$, and these events are disjoint, so

$$\alpha = \sum_{k=0}^{\infty} (a'b')^k a = \frac{a}{1 - a'b'}.$$

Similarly Boadicea wins on shot $2k + 2$ when k rounds pass, Arthur misses and Boadicea hits, with probability $(a'b')^k a'b$. So

$$\beta = \sum_{k=0}^{\infty} (a'b')^k a'b = \frac{a'b}{1 - a'b'} = \frac{(1-a)b}{1 - (1-a)(1-b)}.$$

As a check, $1 - a'b' = a + b - ab = a + a'b$, so $\alpha + \beta = 1$: the contest ends with probability 1.

The contest: expected length

Let T be the number of shots in the contest. From the previous part, $T = 2k + 1$ with probability $(a'b')^k a$ and $T = 2k + 2$ with probability $(a'b')^k a'b$. Writing $q = a'b'$,

$$\mathrm{E}(T) = a \sum_{k=0}^{\infty} (2k+1) q^k + a'b \sum_{k=0}^{\infty} (2k+2) q^k.$$

Ask yourself How do these sums fit the series S?

The first sum is $1 + 3q + 5q^2 + \cdots$, which is S with $r = q$ and $d = 2$. The second is

twice $1 + 2q + 3q^2 + \cdots$, which is S with $r = q$ and $d = 1$. So

$$\sum_{k=0}^{\infty}(2k+1)q^k = \frac{1}{1-q} + \frac{2q}{(1-q)^2} = \frac{1+q}{(1-q)^2},$$
$$\sum_{k=0}^{\infty}(2k+2)q^k = 2\left(\frac{1}{1-q} + \frac{q}{(1-q)^2}\right) = \frac{2}{(1-q)^2}.$$

Hence

$$\mathrm{E}(T) = \frac{a(1+q) + 2a'b}{(1-q)^2}.$$

Now we compare with the target. Using $\alpha = a/(1-q)$ and $\beta = a'b/(1-q)$,

$$\frac{\alpha}{a} + \frac{\beta}{b} = \frac{1}{1-q} + \frac{a'}{1-q} = \frac{(1+a')(1-q)}{(1-q)^2}.$$

So it remains to show that $a(1+q) + 2a'b = (1+a')(1-q)$. Using $1 - q = a + a'b$,

$$(1+a')(1-q) = (1+a')(a+a'b) = a + aa' + a'b + a'^2b,$$
$$a(1+q) + 2a'b = a + aa'(1-b) + 2a'b = a + aa' - aa'b + 2a'b.$$

The difference (first minus second) is

$$a'^2b + a'b - 2a'b + aa'b = a'b\,(a' + a - 1) = 0,$$

since $a + a' = 1$. Therefore

$$\mathrm{E}(T) = \frac{\alpha}{a} + \frac{\beta}{b},$$

as required.

Checks

- If $a = 1$, Arthur hits at once: $\alpha = 1$, $\beta = 0$ and $\mathrm{E}(T) = 1$, matching $\alpha/a + \beta/b$.
- If $a = b = p$, then $\alpha/p + \beta/p = (\alpha + \beta)/p = 1/p$. That is right: the contest is then just a sequence of independent shots, each hitting with probability p, and by the first part the first hit takes $1/p$ shots on average.
- With $a = 0.3$, $b = 0.45$ we get $\alpha = 0.3/0.615 \approx 0.4878$ and $\mathrm{E}(T) = 1.7/0.615 \approx 2.764$; a simulation of 200 000 contests gave 0.486 and 2.769.

Common slips

- Forgetting that Boadicea can only win if Arthur has just missed, and writing $\beta = b/(1 - a'b')$; the missing factor a' shows up when $\alpha + \beta \neq 1$.
- Counting rounds instead of shots: a contest Boadicea wins in round $k + 1$ has $2k + 2$ shots, not $k + 1$.

Beyond STEP

Results like this are usually proved more slickly by conditioning on the first round ("first-step analysis"): if nobody hits in round one, the contest starts afresh after two shots, so $\mathrm{E}(T) = a \cdot 1 + a'b \cdot 2 + a'b'\big(2 + \mathrm{E}(T)\big)$, which gives $\mathrm{E}(T) = (1 + a')/(1 - a'b')$ in one line. First-step analysis is the basic tool for Markov chains, studied in the first or second year of a mathematics degree. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and this style of argument.

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